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F.L.O.W.

Fraud & Loan Optimization Workbench

TEAM-82

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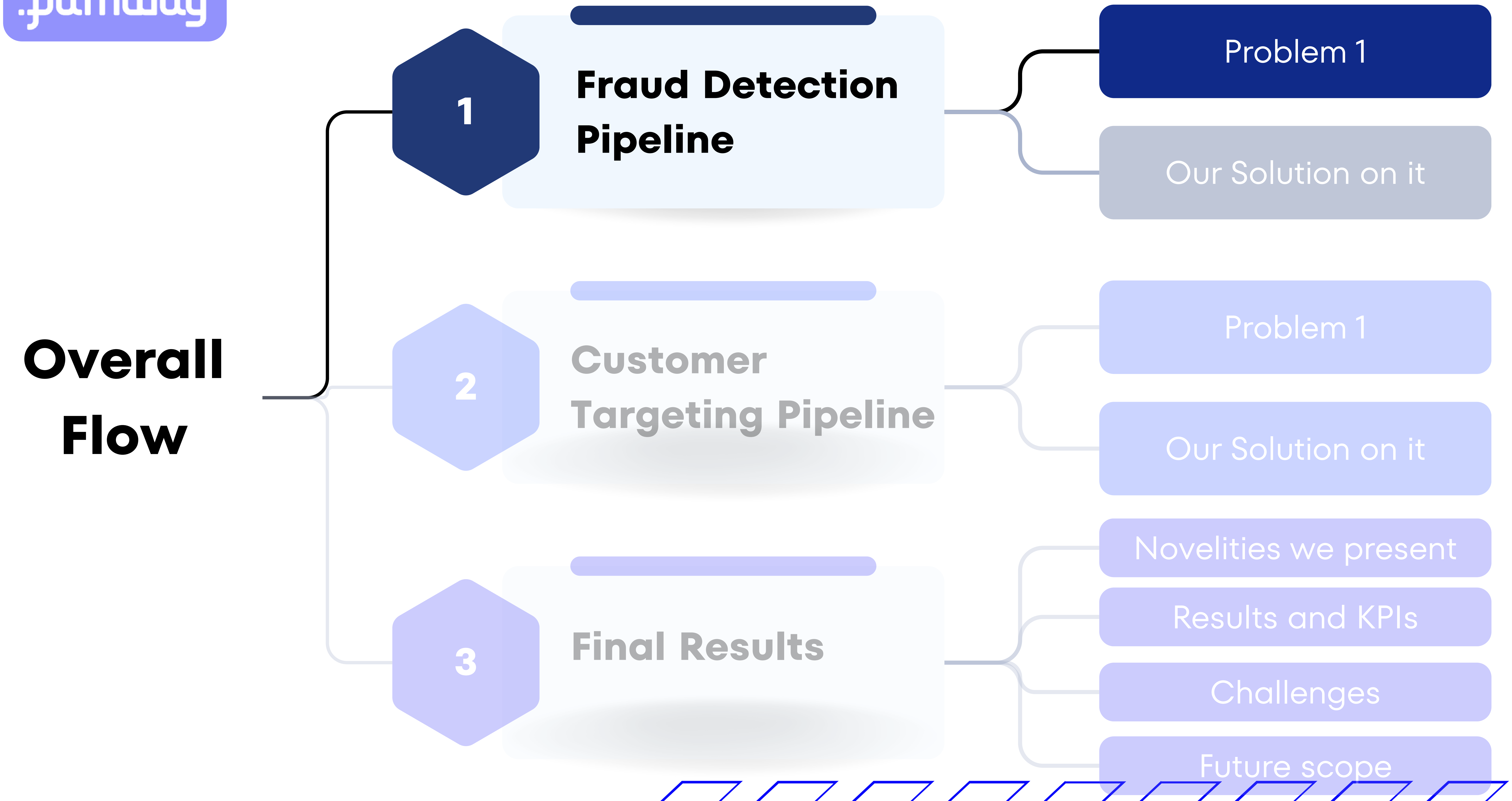
Overall Flow

1. **Fraud Detection Pipeline**
   - Problem 1
   - Our Solution on it
2. **Customer Targeting Pipeline**
   - Problem 1
   - Our Solution on it
3. **Final Results**
   - Novelities we present
   - Results and KPIs
   - Challenges
   - Future scope



Overall Flow
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Fraud Detection Pipeline
Problem 1
Our Solution on it
2
Customer Targeting Pipeline
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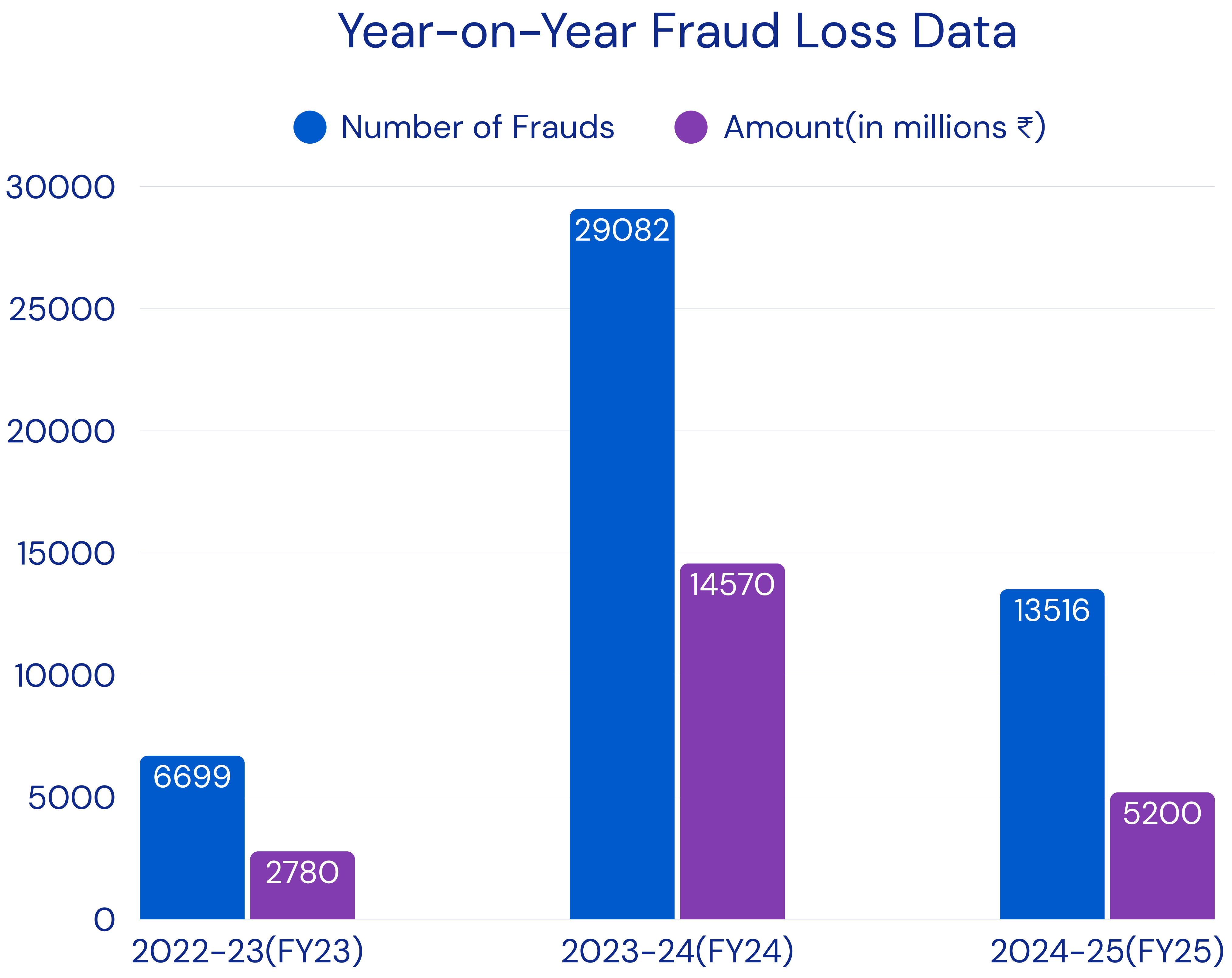
3
Final Results
Novelities we present
Results and KPIs
Challenges
Future scope

PROBLEM-STATEMENT

- Massive fraud cases, huge losses
- Banks also lose money and time on chargebacks, investigations, and losing unhappy customers
- Late detection delays recovery
- Even if banks employ fraud detection methods: they need to be 1) fast 2) accurate 3) adaptable and 4) explainable



Refrence:
https://timesofindia.indiatimes.com/business/india-business/internet-card-frauds-halve-in-fy25-rbi-data/articleshow/121781062.cms

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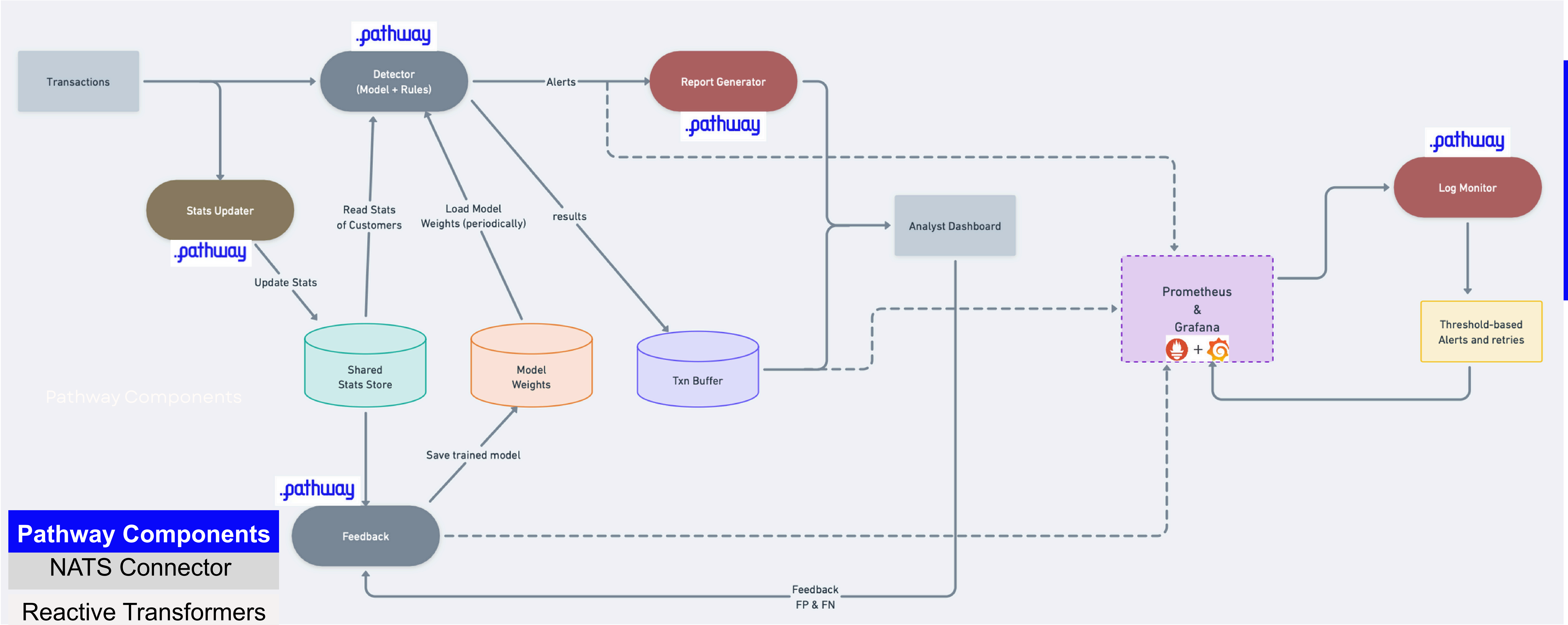
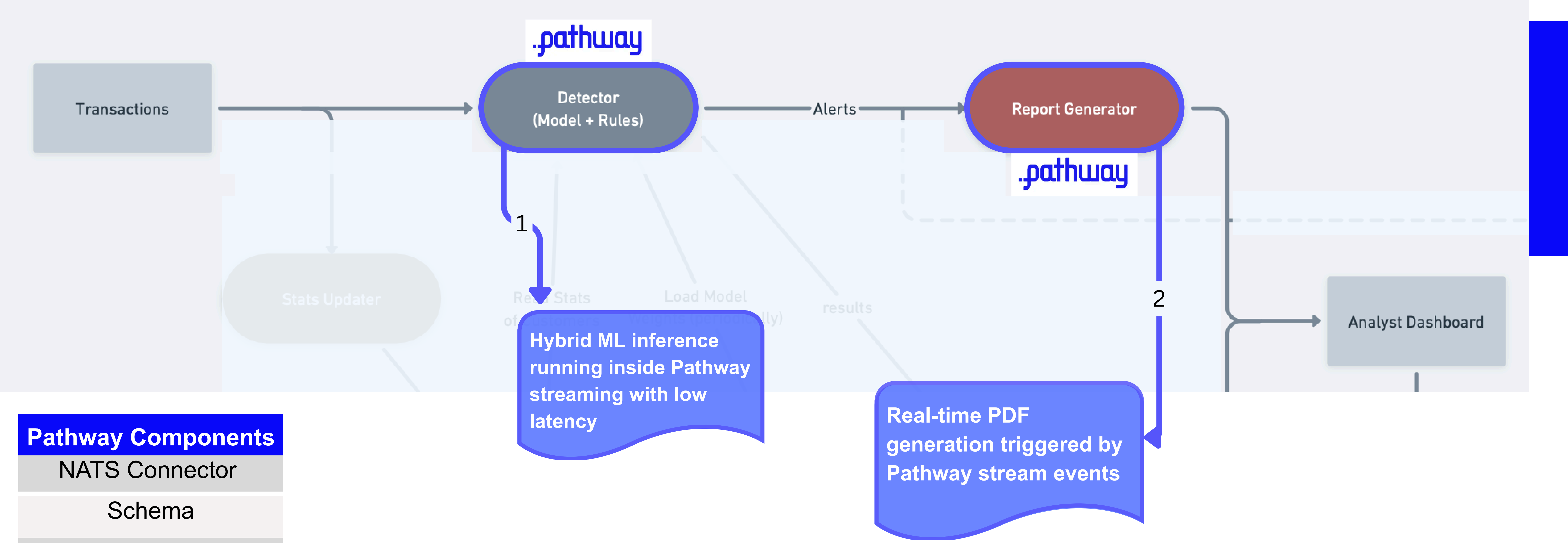
SOLUTION-
PATHWAY POWERED FRAUD DETECTION



Pathway Components
NATS Connector
Reactive Transformers
UDFs
Persistence Layer

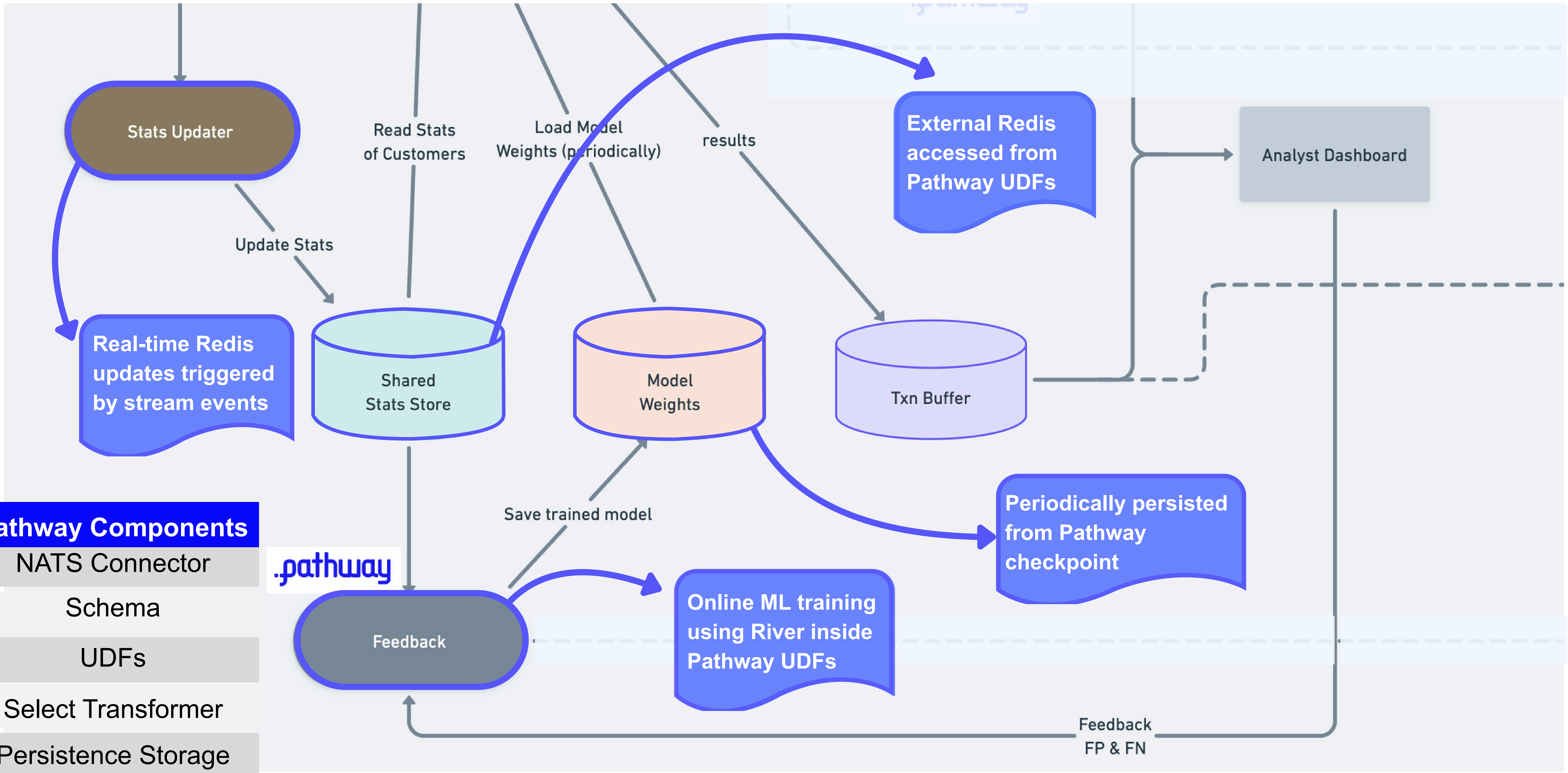
CORE INFERENCE ENGINE



Pathway Components
NATS Connector
Schema
UDFs
Select Transformer
Filter Transformer

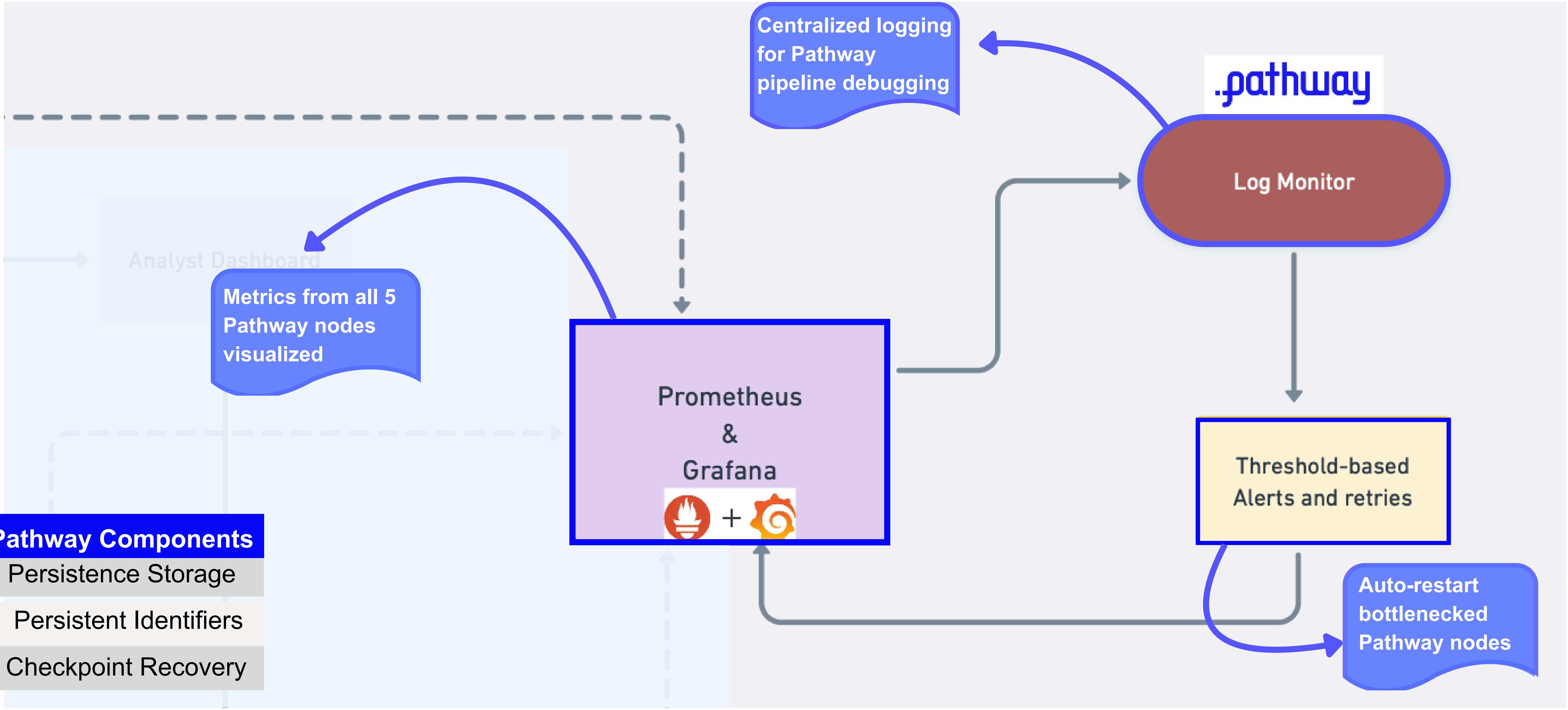
Ref: Adaptive Learning from Evolving Data Streams
Bifet, A., & Gavaldà, R. (2009)

ADAPTIVE LEARNING LOOP



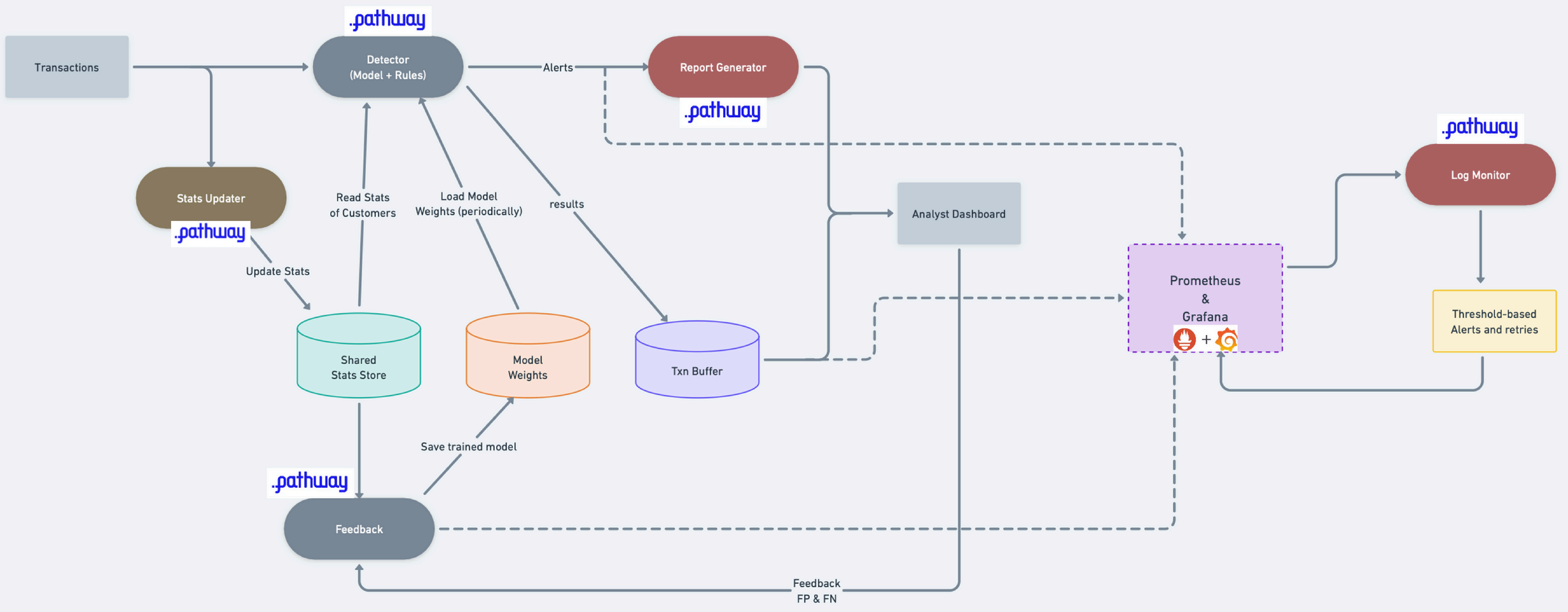
Pathway Components
NATS Connector
Schema
UDFs
Select Transformer
Persistence Storage

OBSERVABILITY AND RESILIENCE

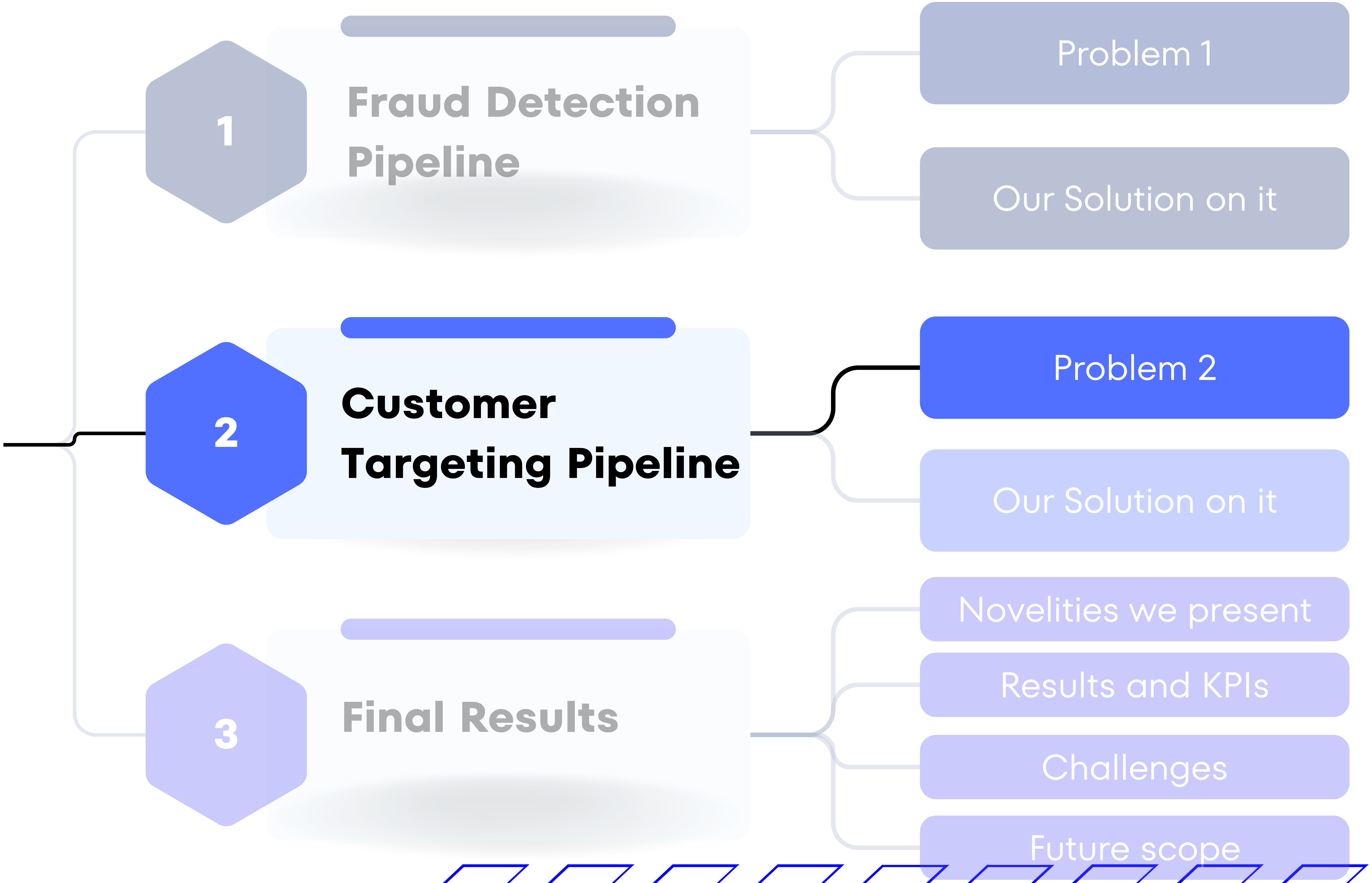


Pathway Components
Persistence Storage
Persistent Identifiers
Checkpoint Recovery

COMPLETE PIPELINE OVERVIEW



Overall Flow



PROBLEM-STATEMENT

- Banks relying on stale CRM
- Loosing Revenue
- Customer Intent evolving real time
- Existing Solutions
- Streaming Native Pipeline
- Highly Likely customers at P.O.T
- Reduced Cold Calls, manpower wastage

Company/Bank	Use Case	IMPACT
Bank of America	Retail Banking	**+19%** Net Income in referenced period
JPMorgan Chase	Cards & Payments	**+30%** digital transactions
Wells Fargo	Engagement	**+500%** in engagements rates
Credit Union of Texas	CRM	**+20%** auto loan leads
PhoneCall.bot	Personalised Calling	**-30%** customer drop-offs

Fig: Growth of Companies from Hyper-Personalisation

Refrence:
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https://www.phonecall.bot/insights/maximizing-commercial-loan-application-submissions-with-ai-phone-calls

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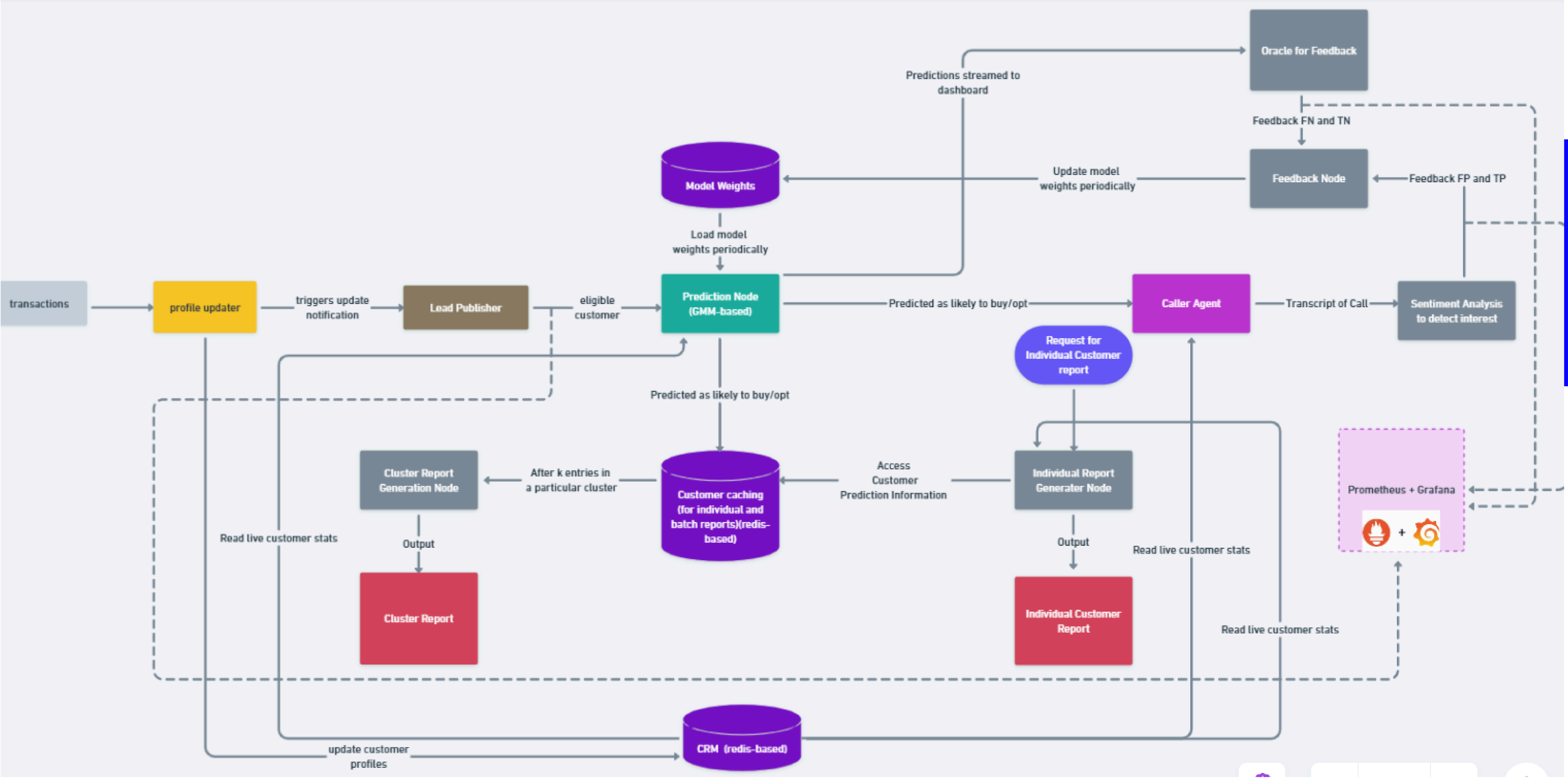
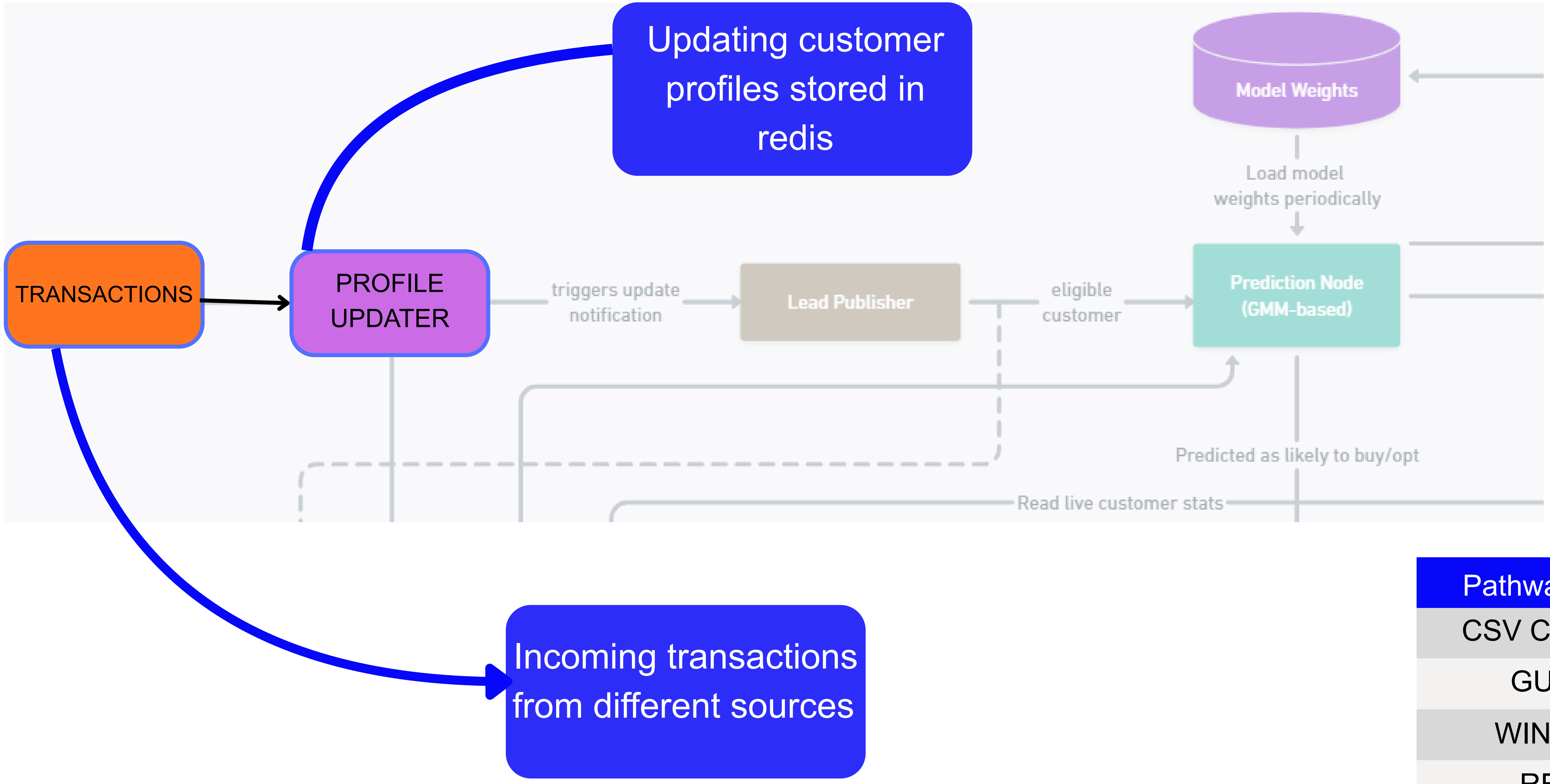
THE PIPELINE

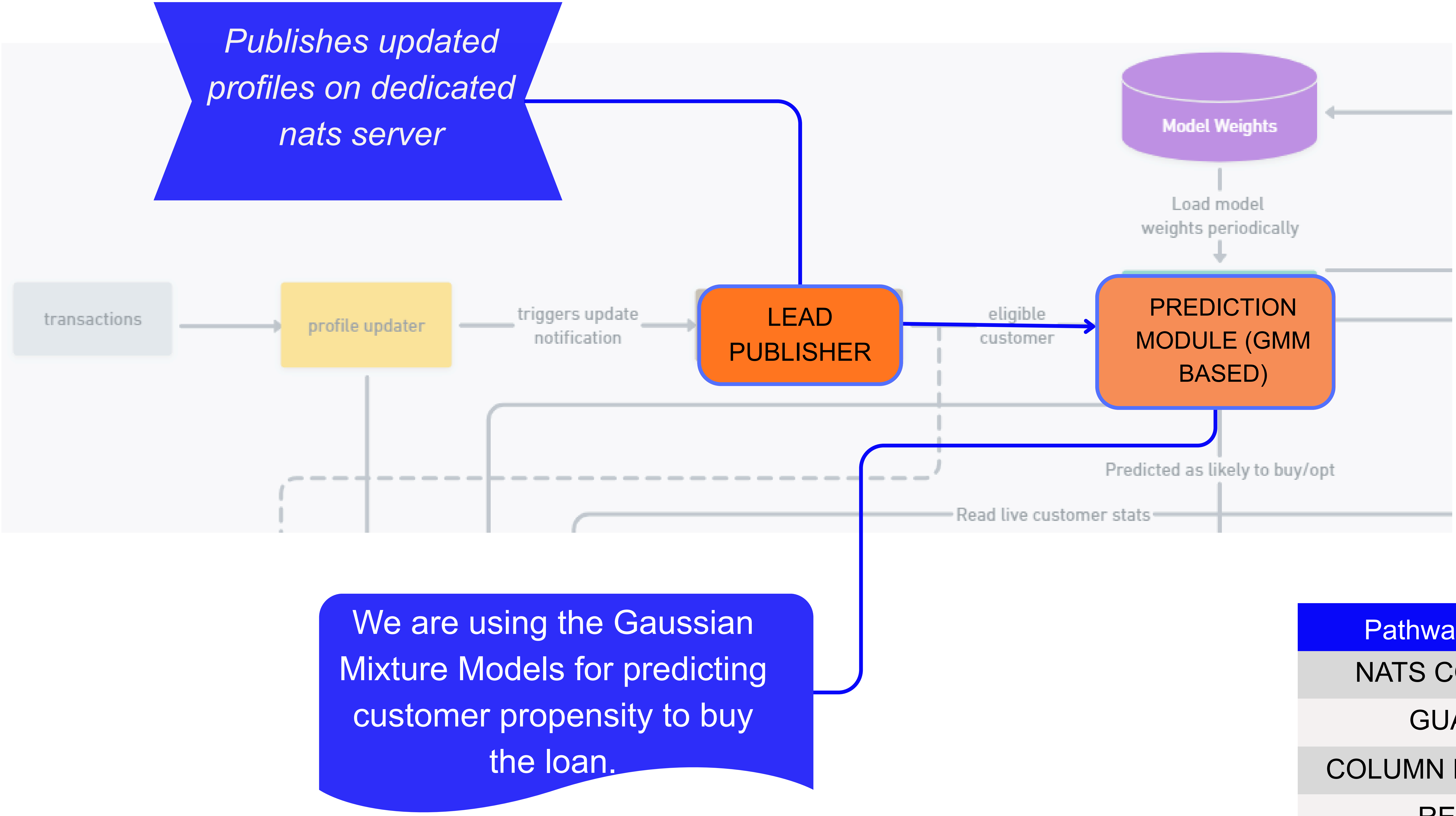


THE STREAMING PIPELINE



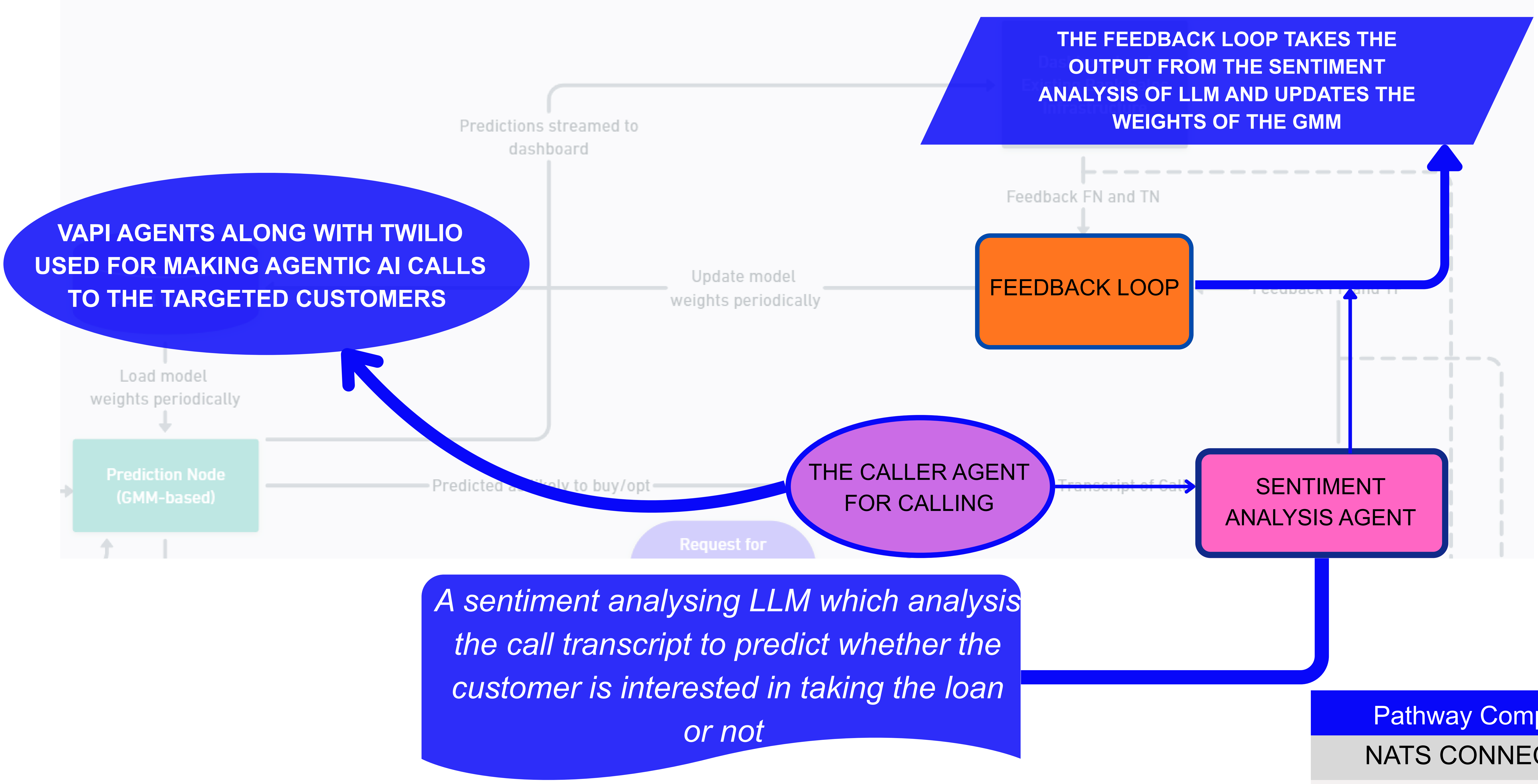
Pathway Components
CSV CONNECTORS
GUARDRAILS
WINDOW JOINS
REDUCERS
FILTERS
SELECT

THE STREAMING PIPELINE



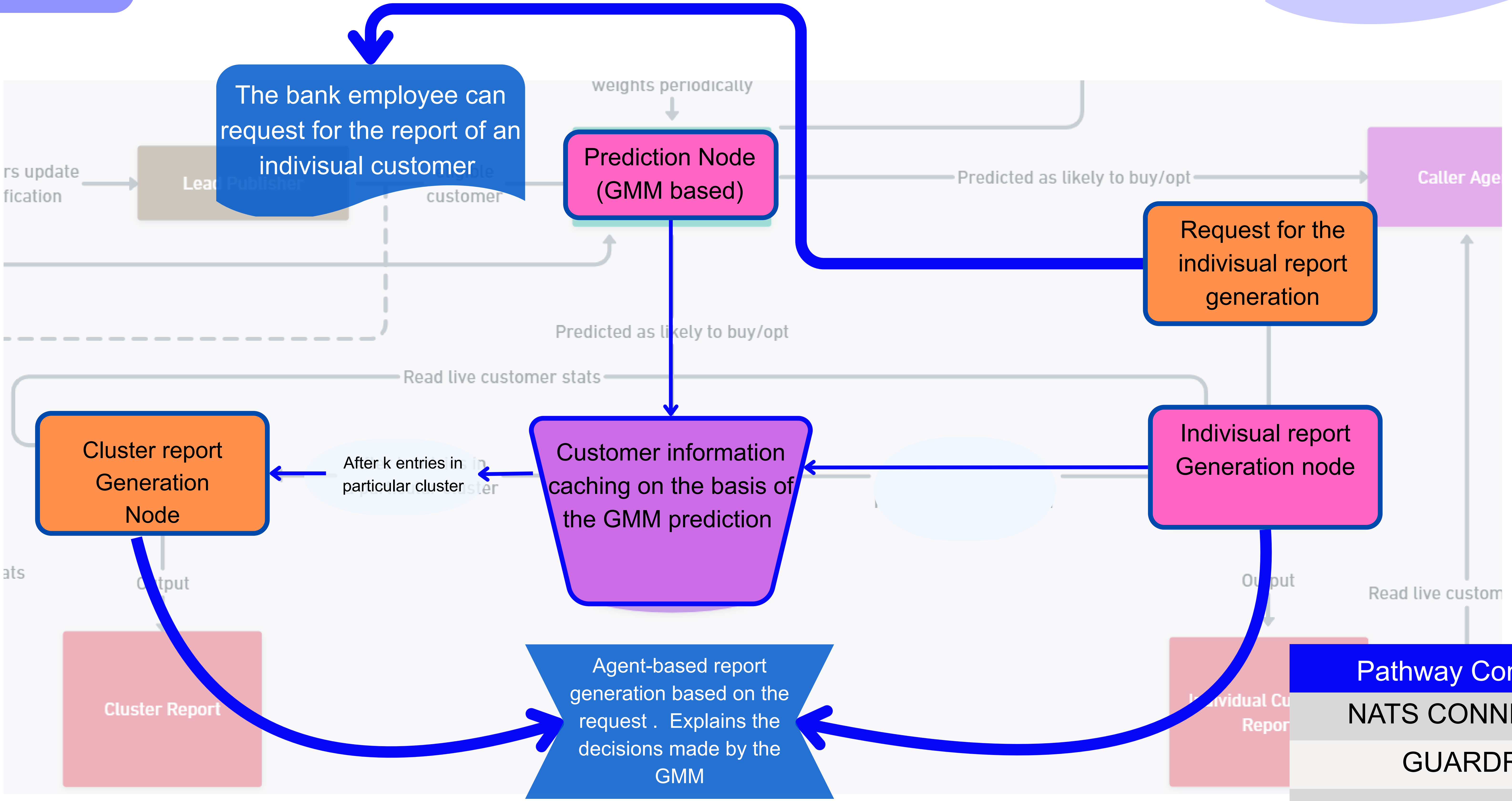
Pathway Components
NATS CONNECTORS
GUARDRAILS
COLUMN EXPRESSIONS
REDUCERS
FILTERS AND SELECT

THE FEEDBACK CALLER PIPELINE



Pathway Components
NATS CONNECTORS
GUARDRAILS
REDUCERS

THE REPORT GENERATION PIPELINE



Pathway Components
NATS CONNECTORS
GUARDRAILS
PATHWAY LLMXPACK
REDUCERS
FILTERS AND SELECT

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NOVELITIES WE PRESENT

Explainable Model Decisions

Many Existing Solutions don’t pay attention to explainable outputs which is an important must in this domain

Adaptive Learning Loop

Delayed feedback updates the model’s weights and penalises wrong decisions

Log Monitoring

Using pathway for log-monitoring for pipeline related alerts/re-starts

DATASET

Customer Targeting

- Customers: 12,500
- Months of Txns: 4
- Total Features: 38

  Income patterns, spending profiles, credit score evolution, DTI ratios, savings behaviour
- Risk signals: bounced payments, high utilization

AI Fraud Detection

- Transactions: 1.9M
- Duration: 24 months (2019–2020)
- Data Source: Credit Card Transactions Fraud Detection Dataset
- Total Features: 22

  merchant categories Travel patterns , fraud-prone segments , Geo-located merchants , movement trails, etc
- Fraud signals: identity fraud, injected patterns, location mismatch, device mismatch

Refrence:
https://www.kaggle.com/datasets/kartik2112/fraud-detection?select=fraudTrain.csv

Results and KPIs

Results of our solutions are given below

Customer Targeting	
Oracle model accuracy	0.88
Oracle model precision	0.89
Oracle model recall	0.97
GMM Accuracy	0.83
GMM Precision	0.86
GMM Recall	0.87

Obtained on our own dataset

AI Fraud Detection	
F1-score	87±2
Latency from publisher to detection	<5ms
Latency from detector to report	<2 sec

Obtained on Credit Card Transactions Fraud Detection Dataset

Refrences:
https://www.kaggle.com/datasets/kartik2112/fraud-detection?select=fraudTrain.csv

CHALLENGES WE FACED

Asynchronous Updates of Model Weights

Both the pipelines needed to update model weights based on the incoming lagged feedback while simultaneously making predictions. We solved this problem by having different nodes for feedback updates and prediction. Both the nodes updated and reloaded common weight files.

Dataset Availability

We did not find reliable public datasets on propensity to buy/opt financial products. We solved this problem by generating the synthetic dataset described previously

FUTURE SCOPE

BDH Integration

We plan to integrate Pathway's new BDH architecture which processes continuous data stream, and adapts without retraining from scratch. So we can use this new architecture to adapt to the explainability of concept drift scenarios in our reports and contact agents without the need for more context or re-training.

Schema Agnostic

It is also possible to extend our pipeline's general idea to other financial scenarios like other investment offerings, and money-laundering detection.

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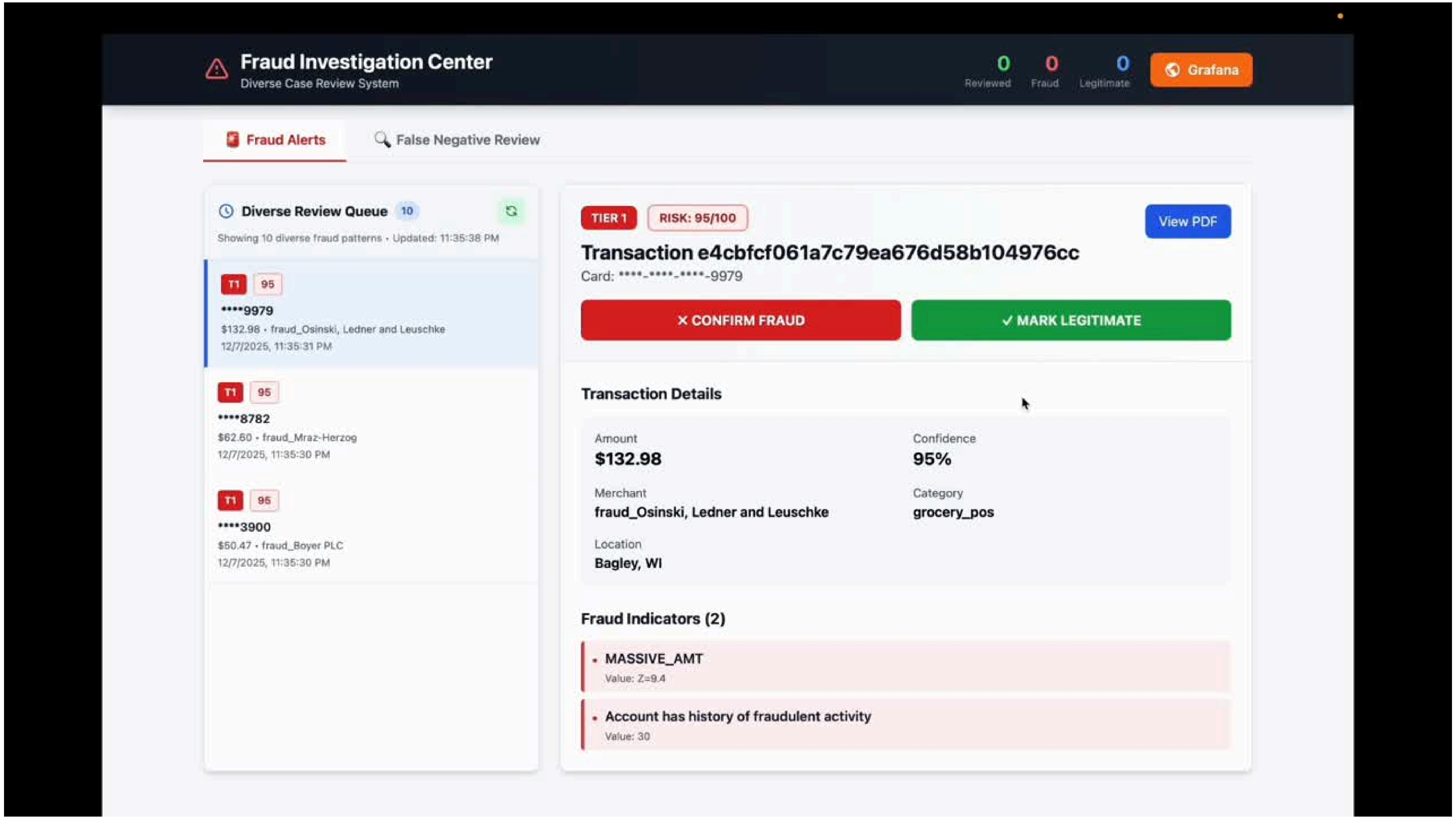
DEMO



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THANK YOU